Topic 3 Computer Hardware

LEARNING OUTCOMES

By the end of this Topic, you should be able to:

1. List four major components of a computer system;
2. Describe the function of central processing unit;
3. Explain various types of computer memory;
4. Define storage;
5. Describe various types of storage systems;
6. Explain various types of input and output peripherals; and
7. Distinguish various types of computer hierarchy.

3.1 INTRODUCTION

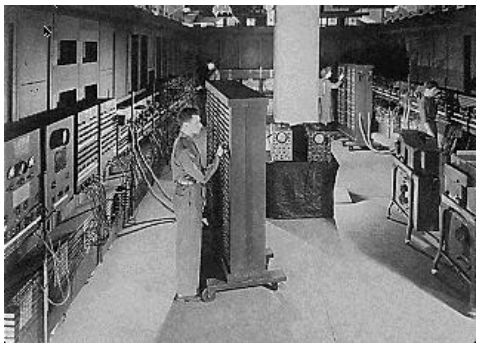


Figure 3.1: The first electronic computer: ENIAC
Source: http://en.wikipedia.org/wiki/ENIAC

In Topic 1 you have learnt about the definition of ICT or computing and the details of it. You were exposed to the architecture and infrastructure of ICT. In Topic 3 you will now learn about computer hardware; which will describe in detail the computer hardware such as the Central Processing Unit (CPU), computer memory, input and output peripherals, and discover the different types of computer hierarchy.

Computer hardware is any component in computing that can be seen and touched. The hardware components are meant to house software to suit a particular activity in businesses or organisations. Users always consider software before hardware selection for a computer system set-up. This reason is that software determine the state or strength of hardware

components. Thus, software applications determine where your computer hardware components are out-of-date or not. The hardware components include system unit and keyboard.

ENIAC *(pronounced /ˈɛni.æk/), short for* ***Electronic Numerical Integrator And Computer****, was the first general-purpose, electronic computer. It was a Turing-complete, digital computer capable of being reprogrammed to solve a full range of computing problems.*

ENIAC was designed to calculate artillery firing (external ballistics) *for the United States Army's Ballistic Research Laboratory, but its first use was in calculations for the hydrogen bomb. When ENIAC was announced in 1946 it was heralded in the press as a "Giant Brain".*

It is important to note that the first effort to create an automatic computing machine in the modern era was carried out by George Babbage, a professor at Oxford University, UK at the end of the 19th century. The Babbage Analytic Machine consists of four components:

i. The storage
ii. The calculating unit
iii. Input unit
iv. Output unit

After ENIAC, there were a number of researchers trying to develop better computers. One of them was John Von Neumann, a professor of mathematics at Princeton Institute of Advanced Studies, who was working on designing a new type of computers called IAS. This computer, which was completed in 1952, is considered to be the prototype of all subsequent general-purpose computers. The architecture of this computer is known as Von Neumann architecture. The development of computers is normally described in term of generations as shown in Table 3.1.

Table 3.1: Computer Generations

Generation	**Major Development**	**Example**
0 (before 1945)	Mechanical computers	Babbage Analytic Machine
1 (1945-1957)	Electronic computers by using vacuum tubes	ENIAC, EDVAC
2 (1958-1965)	Solid state electronics	IBM 7904
3 (1965-1971)	Small and medium scale integrated circuit	IBM System 360, DEC PDP-8
4 (1972-1977)	Large scale integrated circuit Development of microprocessors	IBM System 370
5 (1977- now)	Very Large Scale Integrated Circuit, More powerful microprocessors, Semiconductor memory	PC

3.2 DIGITAL COMPUTERS

In the Von Neumann architecture, a computer system consists of the following components:

i. Central Processing Unit (CPU);
ii. Primary Storage or Main Memory;
iii. Secondary Storage; and
iv. Input and Output Peripherals.

The main role of computer hardware is to do computation. The computation is done by executing a set of instruction (which is normally called a computer program) onto a set of data. Instruction and data are stored in the primary storage of the computer system. During the process of computation, the instruction will be taken by the CPU from the primary storage one by one. If an instruction requires data, then the data will also be taken from the primary storage. Figure 3.2 below shows model of contemporary or smart computer hardware model or structure of a computer system.

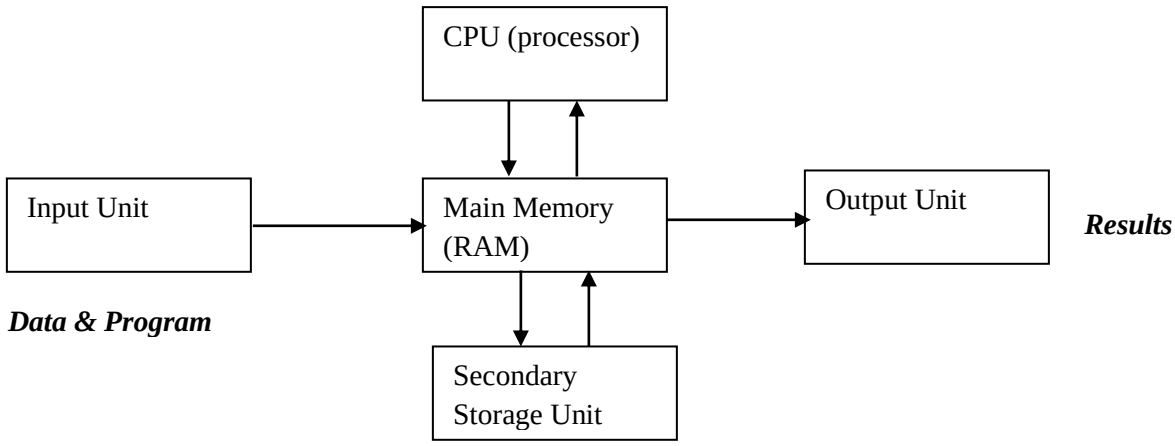


Figure 3.2: Model of Computer Hardware

Since CPU is made up of digital electronic circuit, instructions and data are given in the form of binary codes. So the primary storage must store these instructions and data in this form. Since this type of computer is based on digital technology, it is normally called digital computers or smart computers. The number system that we use in our everyday life is the decimal number system. It is based on decimal digits (0,1,2,3,4,5,6,7,8,9). Digital computer uses the binary number system, which is based only on two digits, 0 and 1. Information or data in digital computers are coded by using a binary code, which is a string of binary digits, for example:

00011010010
11001100111

One binary digit is called a bit and eight bit is called a byte. A binary code that can be

processed at a time by a CPU is called a word. A word length is the number bits in a binary code. Different CPU will have different word length. For example, a CPU with word length of 8 can process 8 bits a time. This type of CPU is called an 8-bit CPU.

3.3 PRIMARY STORAGE

Primary storage is the part of a computer system that is used for storing data and instruction. Primary storage temporary stores data as long as the power is on. It is described as volatile storage, which means its content will erase when the power is turned off. Also, it is called the main storage of a computer. It stores three types of information:

i. Data to be processed by the CPU
ii. Instructions for the CPU
iii. Operating System loaded

3.3.1 Types of Primary Storage

There are two main types of primary storage, namely, RAM and ROM. RAM is an abbreviation for Random Access Memory. This is a type of primary storage that allows its content to be read and edited. In earlier computers, RAMs were made up of an array of ferromagnetic loops referred to as cores. Today, most of the RAMs are in the form of semiconductor chip (See Figure 3.3). Two types of semiconductor RAM, namely:

- Dynamic RAM (DRAM)
- Static RAM (SRAM)

Figure 3.3: An Example of Semiconductor RAM

A conductor is a type of element that can transmit electrical signal. An insulator is the one that resists the transmission of electrical signal. An example of a conductor is gold and copper, while carbon is an example of resistor. Semiconductor is a type of chemical elements whose electrical conductivity is in between that of a conductor and insulator. An example a semiconductor element is silicon and germanium.

Since the 1950s, scientists have found that semiconductor elements can be used to develop electronic components such as transistors, diodes, resistors and capacitors. Later,

semiconductor elements were used in developing integrated circuit that consists of a large number of transistors. The integrated circuit is also known as semiconductor chip (or chip for short). SRAM and DRAM are examples of semiconductor RAM. RAM is temporary and volatile. The contents in RAM will be lost if the electrical supply is turned off.

ROM (Read Only Memory) is a type of primary storage that can retain instructions and data even when the power to the computer is turned off. It is normally used to store instructions that are needed for starting a computer after it has been shut off. In the early days, instruction and data are stored in ROM when the chip is manufactured. However, nowadays, it is possible for us to change the content of ROM. There are a number of different types of ROM. Some of them are:

- PROM (Programmable ROM);
- EPROM (Erasable and Programmable ROM); and
- EEPROM (Electrically Erasable and Programmable ROM).

Apart from RAM and ROM, there is another type of primary storage, which is called cache memory. Cache memory is a high-speed memory that is used to temporarily store blocks of data taken from RAM or ROM. Cache memory is a place closer to the CPU, and thus it is faster than RAM or ROM.

3.3.2 Memory Capacity

Memory capacity is the amount of instructions and data that can be stored in a primary storage. It is measured in term of bytes and one byte is equal to 8 bits. So:

i. 1 Kilobyte (normally written as 1 KB) is approximately equal to 1,000 bytes. In actual fact I KB = 210 bytes.

ii. 1 Megabyte (normally written as 1 MB and read as 1 Meg) is approximately equal to 1,000,000 bytes. In fact 1 MB = 210 x 210 bytes = 220 bytes.

iii. 1 Gigabyte (normally written as 1 GB and read as 1 Gig) is approximately 1,000,000,000 bytes.

iv. 1 Terabyte (normally written as 1 TB and read as 1 Tera) is approximately 1,000,000,000,000 bytes.

3.4 CENTRAL PROCESSING UNIT

The Central Processing Unit (CPU) is the part of a computer hardware that does the actual computation. CPU is a complex electronic circuit. In general, there are two types of CPU:

- CISC (Complex Instruction Set Computer), which provides programmers with many instruction and each instructions is supported by a special purpose circuit.
- RISC (Reduced Instruction Set Computer), which is less complex and hence less expensive than CISC. The drawback of RISC processor is that it places extra demands on programmers, who must consider how to get complex results by combining simple instructions.

3.4.1 Components of CPU

CPU is composed of several distinct parts as shown in Figure 3.3:

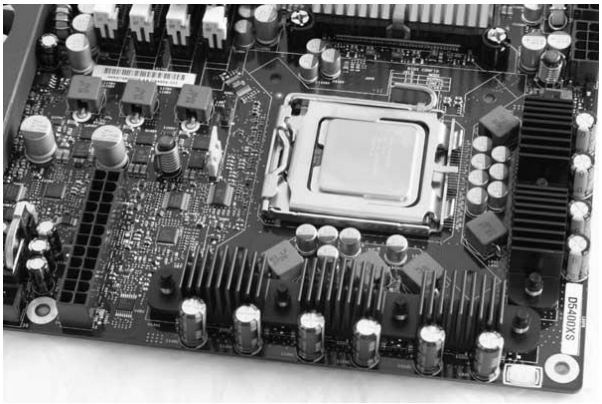


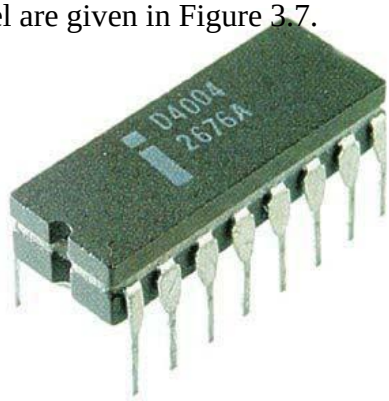
Figure 3.4: Components of a CPU

- Control Unit (CU), which is responsible for controlling the process of computation.
- Arithmetic and Logic Unit (ALU), which is responsible for performing arithmetic and logic operations.
  - Aritmetic Operations: addition, substraction, multiplication, division
  - Logic Operations: and, or, not
- Registers, which are high-speed storage areas that store temporary results and certain control operations. Some of these registers are:
  - Program Counter (PC), which points to the next instruction to be executed.
  - Instruction Register (IR), which holds the instruction currently being executed.
  - Internal registers are registers that can be used to store data that are needed for executing instructions

3.4.4 Microprocessors

In the early days, a CPU is large and is normally stored in a box. Nowadays, a CPU can be manufactured into a single chip. This type of CPU is called a microprocessor. The first microprocessor was produced by Intel Incorporation in 1971, called Intel 4004, which is a 4-bit CPU. These types of microprocessors were used in electronic calculators. Later, Intel produced more sophisticated microprocessors. Examples of microprocessor chips produced by Intel are given in Figure 3.7.

(a)



(b)

Figure 3.5: (a) Intel 4004 (b) Intel Pentium

The development of Intel microprocessors is given in Figure 3.8. As we can see from Figure 3.8, innovation in microprocessor designs has continued to produce more and more powerful microprocessors. The rapid development in the capability of microprocessors had been predicted by Gordon Moore, a co-founder of Intel. In 1965, Moore stated that microprocessor complexity would double approximately every two years. Another interesting feature about microprocessor is that, although its speed and performance has increased drastically, its price has dropped. The drop in the price of microprocessors has enabled computers to be sold at a cheaper price.

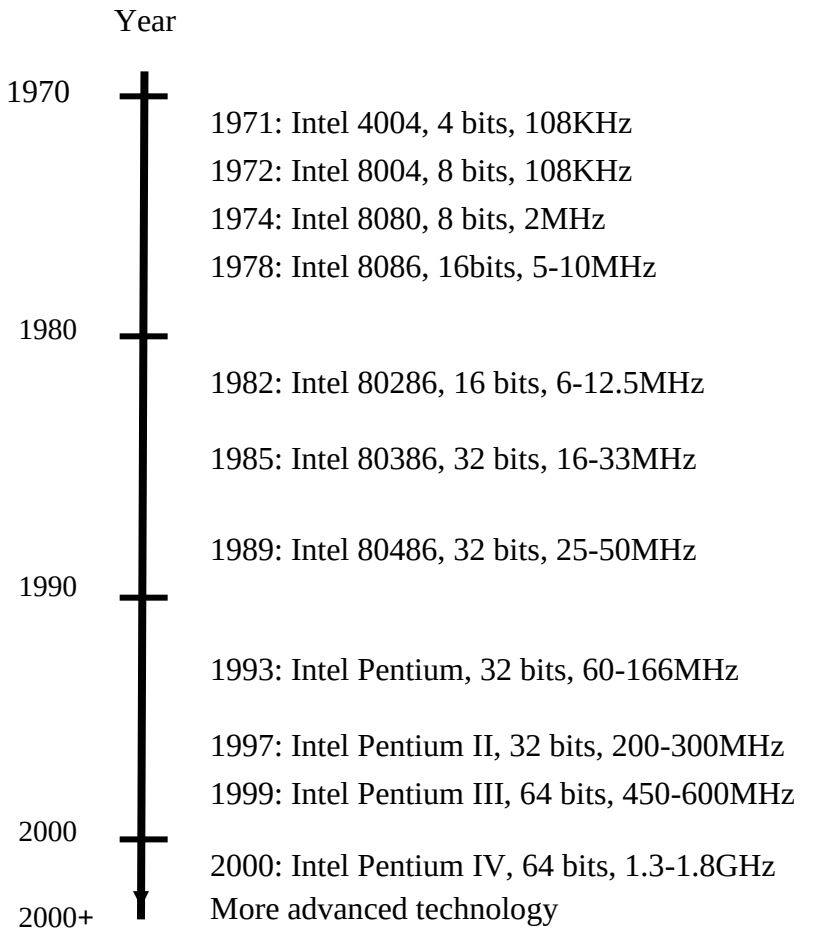


Figure 3.6: Development of Intel microprocessors

3.5 STORAGE DEVICES

Storage device is a medium in computing where files or working data are stored. It is a digital approach or device that preserves files, data or information for retrieval. It can also be defined as a hardware component that stores digital files, data, and information. Examples are videos, audios, photos, and texts.

The main types of storage are primary (see Section 3.3) and secondary. The primary storages are system-oriented that basically handle users' working activities whiles the

power is on. Users have no control over this type of storage, which stores data temporary such as RAM and ROM. The secondary storages are user-oriented. Users have control over such devices such as HDD and Pen Drive.

Also, the storage device or location could be internal or external. The internal includes RAM and HDD. The external includes HDD, OneDrive, Pen Drive, and Google Drive. The internal drives are the main or original storage where files, data or information are basically generated. This means users can by default create their files or works only on the internal storage for routine activities. The external drives are mainly used for backups. Backup is a copy of the original files, data or information. This storage is mainly required to restore the system to its normal routine in case the original is malfunction. The Figure below shows the types of storage devices as of 2024.

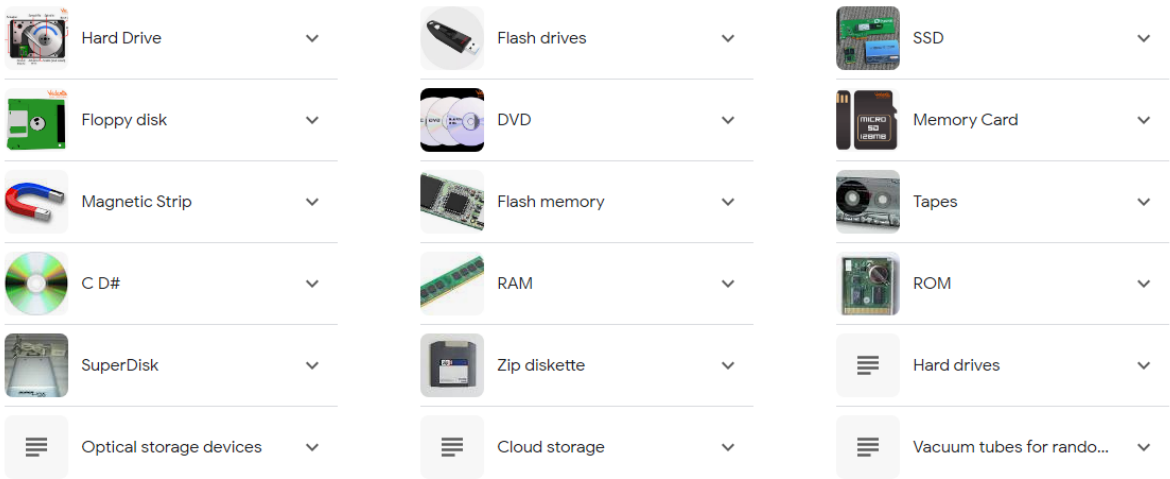


Figure 3.7 Storage Devices/Locations

3.6 SECONDARY STORAGE

The amount of instruction and data that can be stored by primary storage is limited. To store a very large amount of data, most computer systems use secondary storage. Secondary storage is not volatile and is cheaper than primary storage.

There are three different types of secondary storage: magnetic media, optical media and flash memory.

3.6.1 Magnetic Media

Magnetic media store files, data or information via magnetic storage technology. Users use the save or copy command to store their works directly onto these kinds of storage devices. Examples are magnetic tape, magnetic disk, and floppy disk.

Magnetic tape is one of the oldest types of magnetic media. However, it is still being used because it is cheap and can handle a large amount of data. The most popular usage of magnetic tape is to backup information that must be maintained by an organisation. To read information from a magnetic tape, users require special equipment called the tape reader. The picture of a magnetic tape is shown on Figure 3.8.

Figure 3.8: Magnetic tape

To read information from a magnetic tape requires a special equipment called the tape reader. The picture of a tape reader is shown in Figure 3.9.

Figure 3.9: Tape reader

Magnetic disk, called hard disk or hard disk drive (HDD), is the most commonly used secondary storage. It is the popular storage device because of its low cost, high speed, and large storage capacity. It consists of both internal and external storage and operates with 80 tracks. The internal storage or HDD (C:) is the default fixed storage device where all system software and application programmes are stored. The external storage device or drive is movable and meant for backups. Magnetic disks consist of two types, namely;

- Fixed disks or HDD.
- Portable disks or external HDD.

Figure 3.10: Magnetic tape

Floppy disk is another magnetic medium that uses magnetic storage technology to store files, data, and information. It is an external storage device that operates with 80 tracks. Magnetic diskettes, or floppy disks, are similar to external hard disks, except that they are slower and have less capacity. Floppy disks are out of date due to their limited storage capacity to handle multimedia files such as photos and videos.

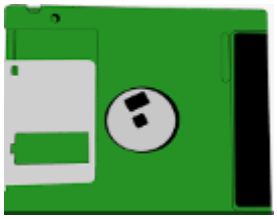


Figure 3.11: Magnetic tape

3.6.2 Optical Media

Optical media store files, data, and information via lasers and lights. Optical storage systems are cheap and can be used to store large amounts of data. This type of storage does not store data via magnetism. Instead, data is recorded by using pinpoint laser beam that burns tiny hole into the surface of a reflective plastic platter. Data is read by another laser installed in the optical disk drive that shines on the surface of the disk. It is also an external storage device that operates with 80 tracks. Users cannot use the save or copy command to store their works directly onto this kind of storage device except via burning the data onto it.

Examples of optical storage devices are:

- CD-ROM (Compact Disk Read Only Memory)

- DVD-ROM (Digital Video Disk Read Only memory)
- FMD-ROM (Fluorescent Multilayer Disk Read Only memory)

Figure 3.12: Magnetic tape

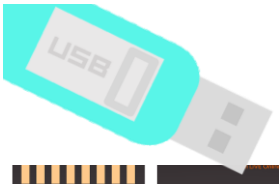
Figure 3.13: Magnetic tape

Users store and read information on optical media via CD/DVD drive. Another technology is CD/DVD rewritable where users can read and write with a writable drive. This technology is also called Blu-Ray optical disc. Users normally read the contents of the optical media in the form of photo, video or audio.

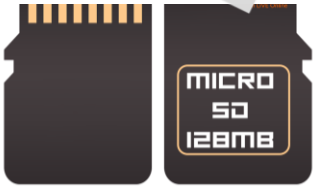
3.6.3 Flash Memory

Flash memory is a type of rewriteable ROM that can be used to store data. This type of storage is normally used in the form of flash card. Users use the save or copy command to store their works directly onto this kind of storage device. This card has only a limited amount of storage capacity. However, it is compact and portable, and is now being used to replace magnetic diskettes and optical storage devices. Examples are Pen Drive, Secure Digital Card, and Memory Card.

Pen Drive (USB) Drive: It is a portable storage device where users store and read information on it via USB port.



Memory Card: is another type of flash memory that stores information in a tiny card with computing devices or computer-embedded devices. The cards are normally found in mobile phones and digital cameras.



Secure Digital Card (SD) Card: It also uses flash memory technology to store files, data, and information on a small device. This technology is an external device. The card is normally inserted into a special slot in laptops.

Figure 3.14: Flash memories

3.6.4 Other Storage Systems

Enterprise storage systems and/or cloud storage systems are public virtual service environment, which are managed by corporate bodies across the globe. These options are external storage systems meant for backups. They are used by many companies or individuals to store large amounts of data. This type provides large amount of storage, high-performance data transfer, protection against data loss and sophisticated management tools. They operate via the Internet for real-time storage and access. Examples are Google Cloud or Drive, IBM Cloud, Amazon Web Services, and Microsoft Azure.

3.7 COMPUTER HIERARCHY

Nowadays, there exist a large number of computers of different sizes and capabilities. These computers can be classified into six categories, name; Supercomputers, Mainframes, Minicomputers, Workstations, Microcomputers, and Computing devices.

3.7.1 Supercomputers

The term supercomputer is normally referred to the type of computer that has the fastest computing engine at a particular time. The most giant of the computer world and it can handle gigantic amounts of scientific computation. It maintained in a special room or environment, may be 50 times faster than a micro, and may cost as much as $20 million. It generally operates four to ten times faster than the next most powerful computer class, the mainframe computer. It is normally run for aerospace, automotive, military and scientific applications, although its use in commercial application has been increasing. Supercomputers are also used for weather forecasting and seismic analysis. It is the giant in the computer world and maintained in a special room or environment. The processor name is called picoprocessor and measured in picoseconds (ps) (one trillion of a second).

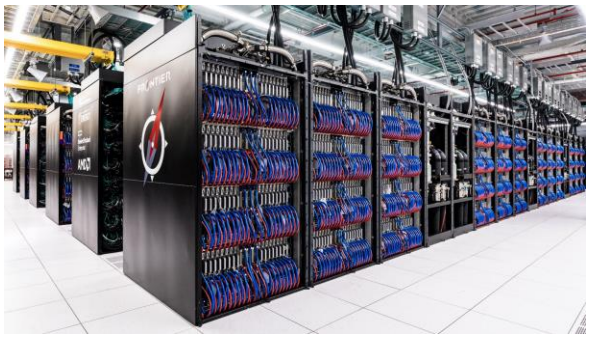


Figure 3.15: Supercomputer

The first supercomputer was produced in 1960 for the Department of Defense, United States. Currently, leaders in development of supercomputers include:

- Cray Research Inc, which produces Cray Supercomputer.
- Thinking Machines Corporation, which produces the Connection Machine.
- Fujitsu, such as Fujitsu FX1.
- Hitachi, such as Hitachi SR2201.

Hitachi's SR2201 series of supercomputers are available in two models: Compact and High-end. The pictures of the two models are shown in Figure 3.16.

(a)

(b)

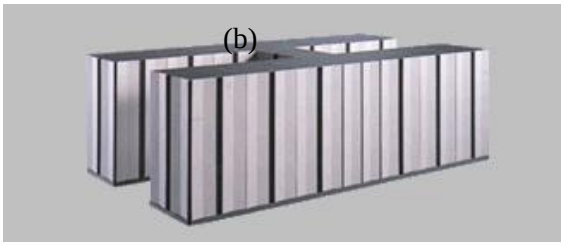


Figure 3.16: (a) SR2201 Compact Model (b) SR2201 The High-end Model

3.7.2 Mainframe Computers

Mainframe computers are widely being used by large enterprises or extensive computing applications that are accessed by thousands of users. Usually housed in a control environment, that can support the processing requirement of hundreds and of ten thousands of users and computer professional. It is smaller and less expensive than a supercomputer Examples of mainframe applications are university student's information systems, corporate payroll systems, large Insurance Companies, a Bank and airline reservation systems. It is mainly used for networking whereby several or individual workstations can be connected.

The first mainframe computer is UNIVAC 1 which was produced in 1951. In the sixties and seventies main manufacturers of mainframe computers were IBM, Burroughs, UNIVAC, NCR, Control Data, Honeywell, General Electric and RCA. Manufacturers outside the USA were Siemens and Telefunken in Germany, ICL in the United Kingdom, and Fujitsu, Hitachi, Oki, and NEC in Japan. IBM was the most dominant manufacturer during this time. An example of mainframe computer is shown in the Figure 3.17.

Figure 3.17: Mainframe computer

In the old days, mainframe computers are connected to hundreds of dumb terminals. Nowadays, most mainframe computers are used as powerful servers connected to hundreds of microcomputers. So, apart from the host processor, a mainframe computer is usually composed of at least two additional processors:

- A front-end processor is responsible for handling communication with all remote computers connected to the mainframe computer system.
- A back-end processor is used to handle data retrieval operation.

The processor name is called nanoprocessor and measured in nanosecond (ns) (one billion of a second). It is normally used for networking whereby several or individual workstations are connected.

3.7.3 Minicomputers

Minicomputers are relatively small and inexpensive computers that perform the same functions as mainframe computers, but with limited capabilities. This type of computer is used to accomplish specific tasks such as process control and for scientific and engineering applications. It can support 2 to 50 computer professional and users. It is mainly used by small and medium-sized companies which can be fit in a corner of a room or a floor next to a desk. The first minicomputers was pdp-1 produced by Digital Equipment Corporation (DEC) in 1957. However, a more popular DEC minicomputers was pdp-11produced in the middle of 1960s.

The current market leader for minicomputers is IBM AS/400. This model was introduced by IBM in October 1978 as a minicomputer for general business and departmental use. In the year 2000, IBM rebranded this product as eServer I Series and then rebranded that again as IBM System in 2006. In 2008, it was again rebranded into IBM Power System and the company is still rebranding the machine according the features. *Another popular minicomputer is Sun Computers which are widely used as e-mail and web servers.* An example of mainframe computer is shown in the Figure 3.18.

Figure 3.18: Mainframe computer

3.7.4 Workstations and Microcomputers

Workstations are powerful single user computers that are normally used to run intensive scientific and engineering applications. Some of these applications are engineering design, medical imaging, scientific visualisation and 3D animation.

Microcomputers are the most popular single user computers. They can be divided into three types:

1. Personal Computer (PC) – Desktops and Towers
2. Thin-client systems
3. Laptop and Notebook Computers

Figure 3.19: Micro computers

Microcomputers were produced since the middle of 1970s and they are:

- Portable
- One processor
- One or single user
- Less cost than all computers
- Need no special room or environment
- Less powerful
- Most likely required on the job

The first successful microcomputers were Apple II. The success of Apple has encouraged IBM to introduce its own microcomputer, called IBM PC in 1981, using Intel microprocessors 8086. The next generation of microcomputer was called IBM PC/AT which used Intel microprocessor 80286. In 1987, IBM decided to stop producing IBM PC. However, other manufacturers continued to produce PC based on the original design of IBM PC. Since then the development of PC follows closely the development of Intel microprocessors, and it is also known based on the type of microprocessors used in the PC. It is portable, one or single user, one processor (microprocessor (mp), most likely required on the job, need no special room or environment, etc

Alternative to PC is Apple computers. One of the most popular types of Apple computers is Apple Macintosh. Apple Computer, IBM and Motorola joined together to develop another type of microcomputer called the PowerPC. Most microcomputers are designed by using the concept of motherboard. A motherboard provides the electrical connections by which the other components of the system communicate. It also hosts the central processing unit, and other subsystems and devices. Modern motherboards include, at a minimum:

- Slots for one or more microprocessors;
- Slots for the system's main memory;
- A chipset which forms an interface between the CPU's front-side bus, main memory, and peripheral buses;
- Non-volatile memory chips containing the system's firmware or BIOS;
- A clock generator;
- Slots for expansion cards; and
- Power connectors, which receive electrical power from the computer power supply and distribute it to the CPU, chipset, main memory, and expansion cards.

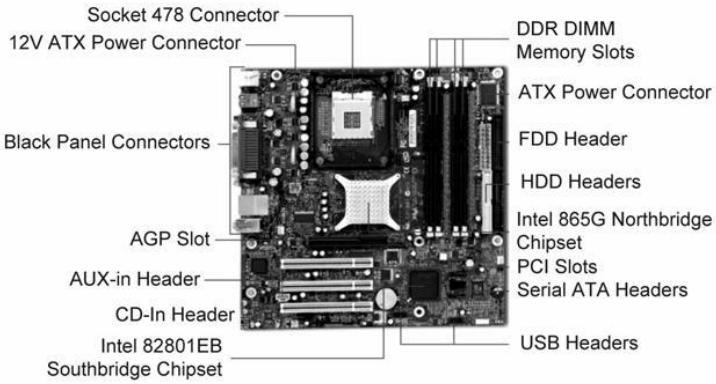


Figure 3.20: An example of a motherboard

3.7.5 Computing Devices

Computing devices refer to smaller and mobile computers. It is a device with computing capabilities. Some of these computing devices are:

- Personal Digital Assistants (PDA)
- Smartphones
- iPads
- Wearable Computers
- Smart Card

Figure 3.21: Computing devices

PDA is also called handheld computers. Most PDAs now are also equipped with Internet access capability. Smartphones are mobile phones which are equipped with some computing capabilities.

Wearable computers are small computers that can be worn and used on the body. An even smaller computer is a smart card. This card contains a small processor and memory that allows it to be used as personal identification card (for example MyKad) and bank card. All debit and credit cards are now produced in the form of smart cards.

Determine which type of computer is suitable for the following activities:

- Controlling a space shuttle;
- Controlling an airplane;
- A meter reader;
- Controlling a scientific experiment;
- Document preparation;
- e-mail server;
- Weather forecast; and
- Teaching of computer programming

3.7.6 Other Devices

IoT stands for **Internet of Things**. It is an interrelated or integrated system of computing devices, mechanical devices, and other objects with unique identifiers (UIDs), which are able to transfer data over a network (wireless) without human interventions. IoT devices are basically not normal computers but they have special components that are internet enabled such as sensors and controllers. More specially, IoT devices are manufactured to work with their specific software applications. Such software applications are normally mobile apps (applications) installed on smartphones with a common protocol for the phones to communicate with IoT devices. This technology includes network between smartphones and TVs, smartphones and vehicles, smartphones and microwave ovens, and smartphones and CCTVs. Figure below illustrates the realty of IoT.

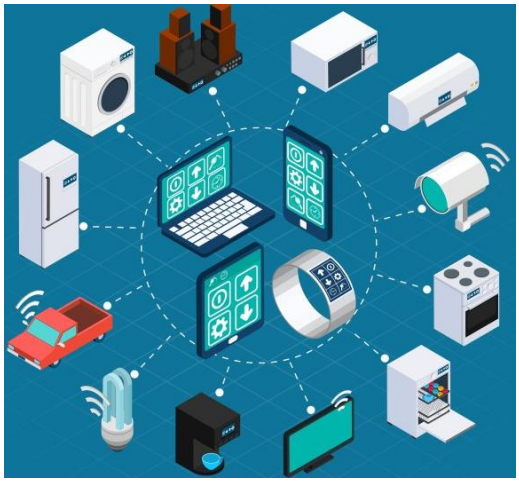


Figure 3.22: Other devices

IoT is also described as home automation system. This technology supports users to control their home appliances, such as how normal remotes control TVs and home theater. Also **IoT Medical Devices** are wearable activity trackers and the most well-known devices. These devices include asthma management, back therapy, and breast health tracking

6.8 INPUT AND OUTPUT PERIPHERALS

Computers communicate with the outside world through input and output peripherals. Two types of input devices are:

(a) Human data entry devices
(b) Source data automation devices

Some of these input devices are listed in Table 3.2.

Type	Device
Human Data Entry Devices	Keyboard
	Mouse
	Joystick
	Touch pad
	Scanner
	Monitor
	Printer
	Plotter
	Touch screen
Source Data Automation	Point of sales terminals
	Barcode scanner
	MyKad Reader
	Optical character recognition
	Sensors
	Camera

Table 3.2: Some input and output devices

The output peripherals include devices such as monitor, printers, plotters and voice output devices. Figure 3.14 shows typical input and output peripherals for a personal computer system.

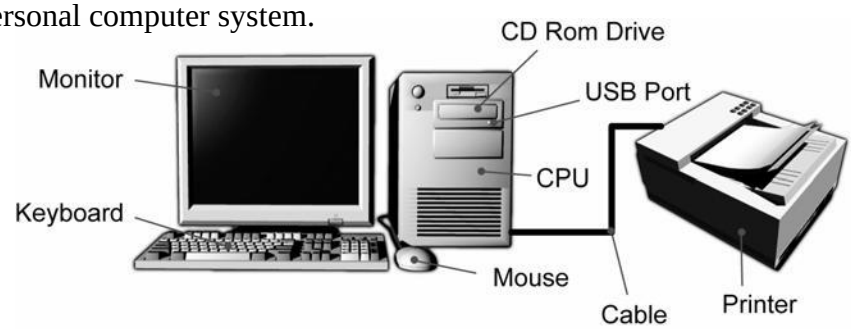


Figure 3.23: A Personal Computer System - Tower

3.9 SELECTING COMPUTER HARDWARE

Selecting computer hardware is a very difficult task since it involves heavy capital investment. The first task in selecting computer hardware is to do a thorough requirement analysis. Since computer hardware technology changes very fast, it is important that the selection is done based on the immediate need of an individual or an organisation.

Types of computer hardware that are normally required can be divided into two categories: individual support and organisational support.
Individual employees would normally need the following items:

- A desktop or notebook PC;
- A printer; and
- A scanner.

In some organisations, a printer is shared between a number of employees. Depending on the types of activities to be carried out, most organisations are now using laser printers since they are capable of producing high quality print output. Since most organisations are now moving towards distributing information electronically, most of them are also providing scanners to some of the employees. By using scanners, hardcopy documents can be scanned and distributed online. Plotters are optional devices depending on the nature of the business of the organisation.

Specification for selecting a computer

1. Processor
2. Hard disk
3. Memory

Mobile devices can be provided some employees, especially those who are always on the move, the mobile devices allow these employees to continue working while on the move.

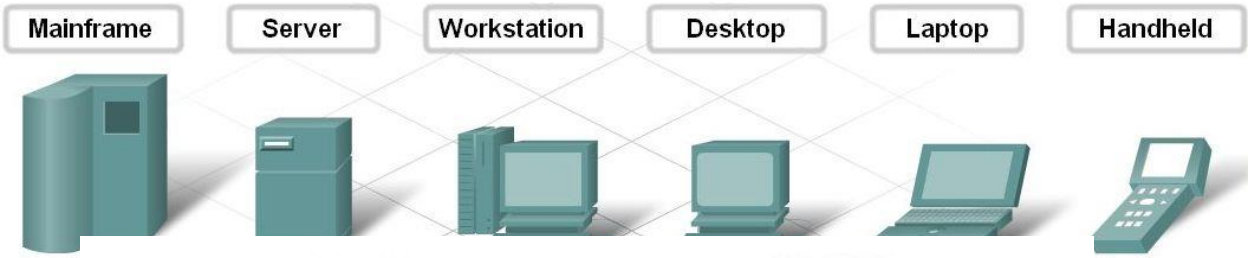
Most organisations need to be equipped with servers. These servers can be classified as:

- Data server
- Mail server
- Web server
- Application server

The type of computers to be used as servers depend on the size of the organisation as well as the amount of activities involved. For example, a small organisation may only require a PC-based server. Bigger organisations may have to use minicomputers or mainframe computers as their servers.

Most organisations buy the hardware needed to support the work for individual employees. However, many smaller organisations choose either leasing or outsourcing for the servers.

Summary

- Computer systems have four major components: the (i) central processing unit (CPU), primary storage or main memory (iii) secondary storage, and (iv) input and output peripherals.
- There are three types of primary storage: (i) Random Access memory (RAM), (ii) Read Only Memory (ROM) and (iii) cache memory.
- Secondary storage can be divided into magnetic media, optical media and flash memory.
- An Enterprise storage system is an independent, external system that includes two or more storage systems.
- Input devices can be divided into two categories: (i) human data entry devices and (ii) source data automation devices. Some of the output devices are monitors, printers, plotters and voice output devices.
- Computers can be divided into six different categories: (i) supercomputers, (ii) mainframes, (iii) minicomputers, (v) workstations, (vi) microcomputers and (iv) computing devices.
- Selecting computer hardware is a very difficult task since it involves heavy capital investment.

Appendix:
Types of Computing Device